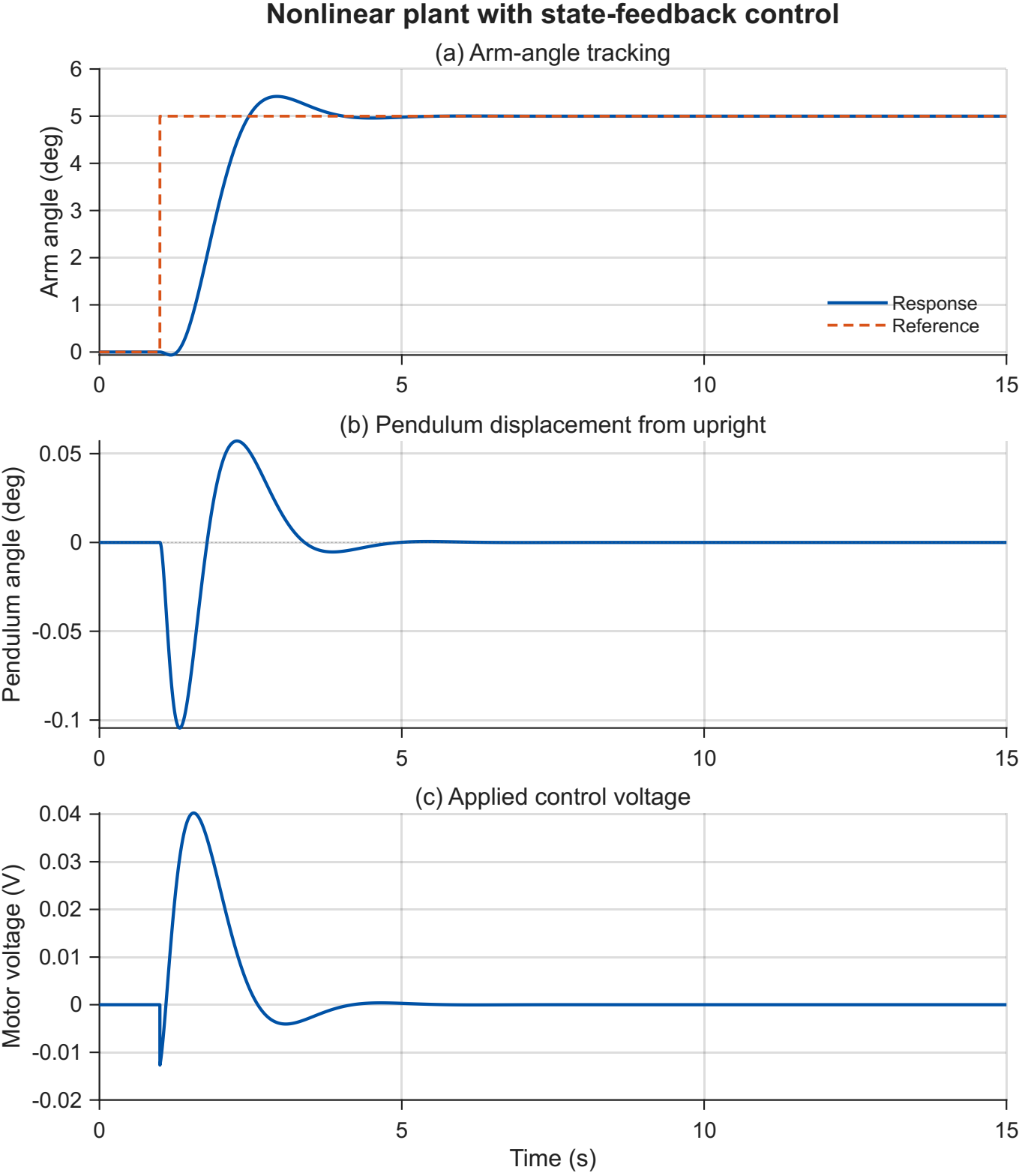

Nonlinear plant with state-feedback control
(a) Arm-angle tracking
Arm angle (deg)
6
5
4
3
2
1
0
Response
Reference
0
5
10
15
(b) Pendulum displacement from upright
Pendulum angle (deg)
0.05
0
-0.05
-0.1
0
5
10
15
(c) Applied control voltage
Motor voltage (V)
0.04
0.03
0.02
0.01
0
-0.01
-0.02
0
5
10
15
Time (s)